

WT EXPERIMENT - 4

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Aim: Component and Arduino Introduction

Introduction:

The Arduino Uno is a versatile and widely used microcontroller board that serves as the foundation for many electronics projects. It is an open-source platform that allows users to design and develop interactive electronic devices. The board consists of an Atmega328 microcontroller that can be programmed to control various hardware components, making it ideal for applications in automation, robotics, IoT, and more.

In this experiment, we will use an Arduino Uno to create an object segregator that differentiates between wet and dry objects using an ultrasonic sensor and a soil moisture sensor. The main goal is to measure the distance of an approaching object, analyze its moisture content, and trigger a servo motor to sort the object accordingly.

Arduino Uno Overview

The Arduino Uno is one of the most popular boards in the Arduino family due to its simplicity, low cost, and ease of use. It features:

- **Microcontroller:** Atmega328P
- **Operating Voltage:** 5V
- **Digital I/O Pins:** 14 (6 can be used as PWM outputs)
- **Analog Input Pins:** 6
- **Flash Memory:** 32 KB
- **Clock Speed:** 16 MHz
- **USB Connection:** For programming and communication
- **Power Supply:** Can be powered via USB or an external power supply

The Arduino Uno can be programmed using the Arduino IDE, which allows users to write and upload code that controls various electronic components connected to the board.

Installation and Setup

Installing the Arduino IDE

1. Download and install the Arduino IDE from [Arduino's official website](#).
2. Connect the Arduino Uno to your computer via USB.
3. Open the Arduino IDE and select the correct board under **Tools > Board > Arduino Uno**.
4. Select the correct port under **Tools > Port**.
5. Upload a sample program to verify the setup.

Running a Test Blink Program

```
void setup() {  
  
  pinMode(LED_BUILTIN, OUTPUT);  
  
}  
  
void loop() {  
  
  digitalWrite(LED_BUILTIN, HIGH);  
  
  delay(1000);  
  
  digitalWrite(LED_BUILTIN, LOW);  
  
  delay(1000);  
  
}
```

If the built-in LED blinks, your setup is working fine.

Components Used:

1. **Arduino Uno:** A microcontroller board used for reading sensor inputs and controlling outputs based on programmed logic.
2. **LED:** A light-emitting diode that can be used to indicate status or provide visual feedback.
3. **Electret Microphone:** A type of microphone that converts sound waves into electrical signals, often used for sound detection.
4. **Resistor:** A passive electrical component that limits or regulates the flow of electrical current in a circuit.
5. **Jumper Wires:** Conductive wires used to connect different parts of a circuit on a breadboard or other prototyping platform.
6. **Breadboard:** A reusable solderless board used for building and testing temporary electronic circuits.
7. **Capacitor:** An electronic component that stores and releases electrical energy, used for filtering, smoothing, or timing in circuits.

Code:

```
const int redLED = 2;
const int yellowLED = 3;
const int greenLED = 4;
const int micPin = A0;

void setup() {
  pinMode(redLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(micPin, INPUT);
  Serial.begin(9600);
}

void loop() {
  int soundValue = analogRead(micPin);
  Serial.println(soundValue);

  if (soundValue > 800) {
    digitalWrite(redLED, HIGH);
    digitalWrite(yellowLED, LOW);
    digitalWrite(greenLED, LOW);
  } else if (soundValue > 500) {
    digitalWrite(redLED, LOW);
    digitalWrite(yellowLED, HIGH);
    digitalWrite(greenLED, LOW);
  } else {
    digitalWrite(redLED, LOW);
    digitalWrite(yellowLED, LOW);
    digitalWrite(greenLED, HIGH);
  }

  delay(100);
}
```

Code Breakdown:

Constant Definitions:

Define pin numbers for LEDs and microphone.

Setup Function:

Set pin modes for LEDs as outputs.

Set pin mode for microphone as input.

Initialize serial communication.

Main Loop:

Read analog value from microphone.

Print the sound value to Serial Monitor.

Conditional logic to control LEDs based on sound value:

High sound level: Turn on red LED.

Moderate sound level: Turn on yellow LED.

Low sound level: Turn on green LED.

Delay:

Pause the loop for a short duration to stabilize readings.

Methodology:

1. Microphone Detection:

The electret microphone continuously listens for sound levels in the environment.

If the sound level exceeds a predefined threshold, the system proceeds to analyze the sound intensity.

2. Sound Level Analysis:

The Arduino processes the analog signal from the microphone to determine the intensity of the sound.

Based on the sound level, the system categorizes the intensity into different ranges (e.g., low, moderate, high).

3. LED Indication:

High Sound Level: If the sound level is very high, the red LED is turned on to indicate a loud environment.

Moderate Sound Level: If the sound level is moderate, the yellow LED is turned on to indicate a medium noise level.

Low Sound Level: If the sound level is low, the green LED is turned on to indicate a quiet environment.

4. Arduino Processing:

The Arduino reads the analog input from the microphone and converts it into a digital value.

It uses conditional logic to determine which LED to light up based on the sound level.

The Arduino provides real-time feedback via the Serial Monitor, displaying the sound values for debugging and monitoring purposes.

Conclusion:

This project demonstrates using Arduino to classify and respond to different sound levels using an electret microphone and LED indicators. It showcases real-time environmental sensing, applicable to noise monitoring and interactive systems.